15 October 1952
Superseding
MIL-A-7292(USAF)
7 February 1952

MILITARY SPECIFICATION

ANTENNA AT-190B/AP

1. SCOPE

1.1 This specification covers one type of antenna, designated Antenna AT-190B/AP.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following specifications, standards, and drawings, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

SPECIFICATIONS:

Military:

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, (General Specification For)

Air Force-Navy Aeronautical:

AN-E-19	Electronic Equipment; General Specification For

U. S. Air Force:

7200	Electronic Equipment; Aircraft, Preservation, Packaging, Packing and Marking For Shipment Of

STANDARDS:

MIL-STD-129	Marking of Shipments

DRAWINGS:

U. S. Air Force:

51D13577	Antenna Base AB-109B/AP Assembly
52D13617	Antenna AT-190B/AP Assembly

Copies of specifications, standards, and drawings required by Contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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3. REQUIREMENTS

3.1 General.- The requirements, including material and workmanship, specified in Specification AN-E-19 are applicable as requirements of this specification with the following additions or exceptions to Specification AN-E-19 ad identified by the similarity of paragraph headings. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.1.1 Finishes.- Aluminum parts which need to be grounded or bonded shall not be anodized. A caustic dip finish is satisfactory. Structural parts which do not need to be grounded or bonded shall be anodized.

3.1.2 Life.- The antenna shall be designed so as to be capable of reliable performance under service conditions specified herein for a total period of 1000 hours.

3.1.3 Moisture and Fungus Resistant Treatment.- Fungus inert materials and components shall be used in the antenna to the greatest extent possible. Overall spray or brushing shall not be used. Any materials or components which must be used that are not in themselves inert to fungus growth shall be treated individually in accordance with instructions from the Procuring Agency.

3.2 Design.- The antenna shall be in accordance with Drawing No. 52D13617 and the following:

3.2.1 Mechanical:

3.2.1.1 Weight.- The weight of the antenna shall not exceed 3 pounds and 12 ounces.

3.2.1.2 Mounting.- The antenna shall be designed to be mounted in the Antenna Base AB-109B/AP in accordance with Drawing No. 51D13577 and held in place by two screws. The Antenna Base AB-109B/AP when installed shall have the underside of the flange against the outside skin of the aircraft.

3.2.1.3 Pressure Seal.- The rubber gasket in the base of the antenna shall make a pressure seal in the Antenna Base AB-109B/AP.

3.2.2 Electrical:

3.2.2.1 Operating Frequency.- The antenna shall cover the frequency range from 90 plus zero minus five to 137 plus five minus zero megacycles per second.

3.2.2.2 Voltage Standing Wave Ratio.- The voltage standing wave ratio in a 50-ohm transmission line feeding the antenna shall not exceed a ratio of 2 to 1 within the frequency range herein specified, with the antenna mounted at the center of a conducting surface at least 10 feet on a side.

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3.3 Power Rating.- The antenna shall be capable of handling the continuous wave r-f power of 200 watts.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of the antenna shall be classified as follows:

4.1.1 Preproduction Tests.- Preproduction tests are those tests accomplished on samples which are representative of the production antenna after the award of the contract to determine that the production antenna meets the requirements of the specification.

4.1.2 Inspection Tests.- Inspection tests are those tests accomplished on the antennas manufactured and submitted for acceptance under the contract.

4.2 Tests Conditions.- The tests shall be conducted under the following conditions unless otherwise specified:

Temperature	Room ambient, 25 degrees Centigrade plus or minus 5 degrees
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity

4.3 Preproduction Tests:

4.3.1 Preproduction Test Sample.- Two antennas as specified in the contract shall be tested for design approval by the Procuring Agency. The acceptance of the samples shall be contingent upon the satisfactory compliance with the requirements for securing the necessary approval for the use of all non-standard parts from the Procuring Agency.

4.3.1.1 Non-Standard Part Samples.- Samples of non-standard parts and materials (and those standard parts listed as Qualification 2 Items in Specification AN-E-19) shall be submitted in quantities and to destinations specified by the Procuring Agency for tests, examination and approval of the agency as required in Specification AN-E-19. Sample parts and materials may be tested to destruction by the Procuring Agency and will not be returned to the contractor. Prior to submission of the non-standard parts for approval for use under this specification, the contractor shall determine that the non-standard parts meet the requirements of this specification, and shall submit with his request for approval sufficient data to establish satisfactory evidence of such compliance.

4.3.1.2 Design Data.- Prior to or at the time of the submission of the preproduction sample antenna, three copies of a report by the contractor of his tests on the antenna, including the preproduction tests except fungus as outlined in this specification shall be supplied together with information concerning the

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failures which occurred and suggestions for improvements in design which might be incorporated in later productions.

4.3.2 Preproduction Test Procedure. - The preproduction test shall consist of the following tests and those specified under the individual tests methods subject to the test conditions specified herein:

4 3.2.1 Altitude. - An altitude test shall be conducted for one cycle only. The pressure change from room altitude and return thereto shall be made as quickly as the chamber construction permits but need not exceed 0.5 inches of mercury per second. The type of chamber used shall be approved by the Procuring Agency.

4.3 2.1.1 Each preproduction sample shall be subjected to the altitude test (3.4 inches of mercury) for one hour. Near the end of this one hour interval, the antenna shall be energized and a power rating test made to determine compliance with 3.3. A thorough visual inspection of the unit shall be made to determine that no damaging arcs were caused by operation at reduced pressure After it has been determined visually that no damage has resulted from the altitude test, the tests specified in 4.4.2 shall be made

4.3 2.2 Humidity - The humidity test shall be conducted in accordance with Specification MIL-E-5272, Humidity Tests, Procedure I, for 15 days, except that the maximum temperature shall be + 50°C. A power rating test to determine compliance with 3.3 shall be repeated each day when the temperature is 38°C or less. This test shall be followed by the fungus test. The type of chamber used shall be approved by the Procuring Agency.

4.3 2.3 Fungus. - After the antenna has been subjected to the humidity test and has sufficiently dried out, it shall be subjected to the Fungus Resistance Test, Procedure I of Specification MIL-E-5272. At the end of the test, the parts shall be visually examined for evidence of fungus growth. There shall be no evidence of fungus growth which in the opinion of the Procuring Agency will be detrimental to the performance of the equipment

4.3 2 4 Vibration. - At the termination of the vibration test, in accordance with Specification AN-E-19, a thorough visual inspection of the complete antenna shall be made to detect any damage caused by vibration After the above visual inspection, the test specified in 4.4.2 shall be repeated, to determine that there has been no degradation in the performance of the antenna below the requirements of this specification as a result of the vibration test.

4.3.2.5 Shock. - Compliance with the shock requirements shall be checked in accordance with Specification MIL-E-5272, Shock Tests, Procedure II. At the termination of the shock test, the tests specified in 4.4.2 shall be repeated to determine that there has been no degradation in the performance of the antenna below the requirements of this specification as a result of the shock test.

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4. 3. 2. 6 Electrical. - The voltage standing wave ratio shall be measured to determine compliance with 3. 2. 2. 2 at intervals of approximately 10 megacycles and a frequency range from 90 megacycles to 137 megacycles.

4. 4 Inspection Tests:

4. 4. 1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of the tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all of the requirements and tests in the specification and acceptable to the Government.

4. 4 2 Individual Tests. - Each antenna shall be subjected to the following tests:

4. 4. 2. 1 Mechanical:

4. 4. 2. 1. 1 Visual. - A visual inspection shall be made to verify that the materials, design, construction, necessary mechanical measurements, marking, and workmanship comply with the requirements of this specification.

4. 4. 2. 2 Electrical. - The voltage standing wave ratio, using measuring equipment approved by the Procuring Agency, shall be measured to determine compliance with 3. 2.2. 2 for frequencies of 90 megacycles, 110 megacycles and 137 megacycles.

4 4. 3 Sampling Tests - Sampling Tests in accordance with the test procedure set up for the preproduction tests and shall be performed to the extent considered necessary by the Government Inspector on approximately the following quantity.

Quantity of Item on Order	Quantity of Items to be Tested
1 to 25	1
26 to 50	2
51 to 75	3
76 to 125	4
126 to 200	5
201 to 300	6
Over 300	1 for each additional 100 on order

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4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification and drawings, to determine compliance with this specification and drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order

5. PREPARATION FOR DELIVERY

5.1 General. - The antenna shall be prepared in accordance with Specification No. 7200, Method II.

5.2 Marking of Shipments. - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129 The nomenclature shall be as follows: Antenna AT-190B/AP, Specification MIL-A-7292A.

6. NOTES

6.1 Use. - The antenna covered by this specification is a vertically polarized, streamlined stud antenna. The frequency range of the antenna is 90 to 137 megacycles per second.

6.2 Ordering Data - Invitation for bids, contracts and purchase orders should state the conditions for the following:

6.2.1 Preproduction Tests. - It is expected that the contract or purchase order will specify that two antennas shall be required as the First Article and that the First Article shall be subjected to the Preproduction Tests to determine compliance with the requirements of the specification, and that the pertinent data taken by the contractor shall accompany the First Article. The invitation for bids and the contract shall specify the point of inspection for these tests

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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